Fiscal Year:	FY 2013	**Task Last Updated:**	FY 05/23/2013
PI Name:	Yaswen, Paul Ph.D.		
Project Title:	Epigenetic effects of radiation on epithelial cell self-renewal		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	P_Yaswen@lbl.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	510-486-4192
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	University of California		
PI Address 2:	1 Cyclotron Road		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-8099	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
Start Date:	03/01/2010	**End Date:**	02/28/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Straume, Tore	**Contact Phone:**	650-604-3943
Contact Email:	tore.n.straume@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Interagency agreement with DOE (per ARC)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Parvin, Bahram (Lawrence Berkeley National Laboratory) Kronenberg, Amy (Lawrence Berkeley National Laboratory)		
Grant/Contract No.:	NNA10DE03I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Aberrant self-renewal is a fundamental property of cancer cells that can be conferred by non-mutational (epigenetic) means. Early events in radiation-induced carcinogenesis must either cause the expansion of pre-existing cells with extensive self-renewal potential or the acquisition of extensive self-renewal potential by cells that have repressed it. A current weakness in cancer risk assessment models is the lack of consideration of epigenetic factors that influence malignant transformation. Recent data from our own lab as well as others indicate that radiation can promote changes in the epigenetic properties of cultured human cell populations and intact rodent organs. Using a three-dimensional cell culture system that supports tissue-like behavior of human breast cells, we will: 1) determine the effects of ionization density and dose on the frequency of self-renewal versus differentiation, 2) identify biochemical pathways that show persistent changes as functions of ionization density and dose, and 3) use genetic manipulations to test the mechanistic involvement of discrete pathway components in persistent radiation quality effects. These experimental studies will employ new assays to improve the mechanistic understanding of radiation quality effects on the gain of aberrant self-renewal capacity and loss of responsiveness to differentiation signals--two related cancer development processes. The intent of this project is to develop a breast tissue-specific risk model of major processes in radiation carcinogenesis. Enhanced self-renewal and loss of differentiation are important surrogate endpoints that can be used directly in estimates of cancer risk and in reducing the uncertainties in predicting radiation quality effects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are developing assays of aberrant breast cell differentiation and self-renewal. These important surrogate endpoints will be useful in better estimating cancer risk associated with radiation and other potential carcinogens.
Task Progress:	To avoid selection bias associated with expansion in adherent cultures, we performed irradiation experiments using thawed primary epithelial cell-enriched human breast tissue fragments (organoids) in suspension. Our general scheme involved the following steps: 1) overnight incubation of thawed organoids in serum-free medium, 2) acute irradiation, 3) overnight recovery, 4) dissociation to single cells, 5) growth and differentiation assays. In initial experiments, we detected dose-dependent increases in the percentages of “side-population” cells present after irradiation with up to 5 Gy of X-rays. Side population cells identified using this technique are those cells that do not retain the DNA-binding dye, Hoechst 33342, usually because they express high levels of the ATP-binding Cassette Transporter ABCG2 that actively pumps out the Hoechst dye. This assay is one of the oldest and most trusted techniques for identifying and sorting stem and early progenitor cells in a variety of tissues and species. During the spring 2011 run (NSRL11A), organoid aliquots from two specimens were exposed to 50 and 100 cGy doses of gamma rays, as well as 25 and 50 cGy doses of Si and Fe. After overnight recovery, the control and irradiated samples were dissociated to single cells, and aliquots shipped back to LBNL for side population assays. Changes in proportions of SP cells were noted, but no consistent pattern emerged. Because of uncertainties inherent in the shipment of live cells cross-country for analysis, this assay was abandoned in favor of assays that could be performed onsite at BNL, or after fixation. The first such assay involved staining and sorting for c-kit, a cell surface marker that others have shown to be highly expressed by luminal estrogen receptor negative progenitors in both mouse and human mammary glands. We found that ~ 1% of the cells within unirradiated organoids expressed c-kit, and that the level was fairly constant in irradiated samples, with the exception of one set of samples irradiated with the highest dose (0.5 Gy) of Fe ions, where the percentage of c-kit(+) cells was significantly decreased. Beginning with the fall 2011 run (NSRL11C), we examined the effects of radiation on a second set of cell surface markers, CD44 and CD24. The fraction of CD44hi/CD24Lo cells has proven useful for distinguishing stem-like cells in both malignant and non-malignant human breast tissues. To evaluate the possibility that the trypsinization procedure itself would alter the relative abundance of CD44hi /CD24Lo cells, we subjected aliquots of the same organoid preparation to trypsin for differing amounts of time. The results demonstrated that trypsinization for periods between 5 and 20 minutes did not lead to differences in the fractions of CD44hi/CD24Lo cells, confirming that the dissociation procedure itself was not a selective influence. Results obtained with organoids from several individuals, each run in triplicate, indicated that the baseline fraction of cells with this phenotype was relatively stable for each specimen, but ranged between 0.5 and 5% among specimens. During the fall 2011 run (NSRL11C), organoid aliquots from two specimens were exposed to 50 and 100 cGy of gamma rays, 25 and 50 cGy doses of Si (300 MeV/amu), as well as 10 and 25 cGy doses of Fe (300 MeV/amu and 1 GeV/amu). The results indicated that, at the low to moderate doses of radiation employed, no significant changes in the fractions of CD44hi/CD24Lo cells were detectable 24 hours after irradiation, regardless of ion or charge density. To assess the differentiation potential of primary human mammary epithelial cells (HMECs) derived from control or irradiated organoids, 24 hours after irradiation the organoids were dissociated with trypsin to single cells. The isolated cells were then incubated at clonal densities in adherent cultures for 7 days before fixation and staining with specific antibodies for keratins 14 (basal marker) and/or 19 (luminal marker). In preliminary experiments, when irradiated with 1 Gy of X-rays, the percentages of mixed colonies containing both luminal and basal cells (indicating those colonies likely to have been generated from multipotent progenitors) were significantly higher for one specimen and marginally higher for another specimen, than for corresponding unirradiated controls. However, during the fall 2011 run (NSRL11C), when organoid aliquots from different specimens were exposed to the ions and doses indicated above, the specimens did not show statistically significant differences in the percentages of mixed colonies generated from unirradiated versus irradiated organoids. Subsequent exposure of organoids from one of these latter specimens to 1 Gy of X-rays at LBNL also did not yield statistically significant differences in the percentages of mixed colonies. The variability observed was therefore likely to be attributable to heterogeneity among the specimens themselves, rather than to beam characteristics or uncontrolled technical issues. During the fall 2011 run, we also assessed the overall colony-forming efficiency of individual cells from a single specimen (N158) that were irradiated as organoids, maintained as organoids for 24 hours post-irradiation, and dissociated to permit single cells to form colonies. Organoids from the same individual were exposed to gamma rays or 1 GeV/amu Fe ions. The results demonstrated that graded low doses of gamma rays do not elicit measurable toxicity in this standard assay, suggesting that when left intact, the local physical environment exerts a protective effect on potential clonogens for our sparsely ionizing reference radiation. In contrast, exposure to graded low doses of 1 GeV/amu Fe ions does result in increasing cytotoxicity, despite the integrity of the local physical environment.

	Beginning with the spring 2012 run (NSRL12A), we examined the effects of radiation on another set of cell surface markers, MUC1 and Thy1, that has proven useful for distinguishing different types of colony-forming cells (CFCs) in non-malignant human breast tissues. These CFCs share expression of EPCAM and CD49f, but luminal-restricted CFCs express MUC1, while less differentiated bipotent CFCs express Thy1. We determined, using specimens from several individuals, each run in triplicate, that the proportion of luminal-restricted CFCs increased, while that of bipotent CFCs decreased in organoids exposed to moderate levels of sparsely ionizing X-rays and gamma rays. However, this correlation was not observed for organoids exposed to approximately equitoxic doses of Fe (300 MeV/amu and 1 GeV/amu). The results suggested possible qualitative differences in the epigenetic responses of progenitor cells in intact glandular structures to the effects of sparsely and densely ionizing radiation. However, this distinction did not hold up in experiments performed with additional specimens during the fall 2012 run (NSRL12C). In summary, over the course of 3 years, we conducted a variety of differentiation and self-renewal assays with primary human organoid preparations exposed to sparsely and densely ionizing radiation. While radiation-associated changes were observed in some cases, no definitive trends were documented. We speculate that radiation-induced changes may have been affected by pre-existing inter- and intra-individual cellular heterogeneity in the organoid preparations. This heterogeneity may have had technical causes (e.g., differences in preparation or handling of specimens), and/or it may represent true biological variation (e.g., differences in position within the originating tissue or genetic variability among specimen donors).
Bibliography Type:	Description: (Last Updated: 05/28/2013)
Abstracts for Journals and Proceedings	Yaswen P, Kaur G, Gauny S, Parvin B, Kronenberg A. "Epigenetic effects of radiation on epithelial cell self-renewal." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012